

HYBRID FOURIER NEURAL OPERATORS FOR TIME-DOMAIN BLACK-HOLE PERTURBATIONS AND QUASINORMAL MODE EXTRACTION

EXECUTIVE SUMMARY

Jonathan Chung | MPhil Data Intensive Science | University of Cambridge

Motivation & Scientific Justification

A perturbed black hole settles by emitting a ring-down signal composed of damped oscillations, the quasinormal modes (QNMs). For each mode of a Kerr black hole, general relativity fixes the frequency and damping time through the remnant mass and spin [1,2]. Measuring these tones in gravitational-wave data tests whether the remnant is consistent with the Kerr prediction, an observational programme opened by the ringdown analysis of GW150914 [3]. Reliable spectroscopy requires an accurate time-domain waveform and a fit interval in which the QNM model is valid [4].

Physics-informed neural networks (PINNs) approximate differential equations by minimising their governing-equation, initial-data and boundary residuals. Patel et al. [5] solve the Regge–Wheeler and Zerilli equations in the time domain, but report relative L^2 field errors of 23.59% and 28.06%, respectively, and select one fit window manually. We first reproduce and improve this calculation. We then introduce separately trained Schwarzschild and Kerr hybrid Fourier neural operators (hFNOs) that correct reduced-resolution numerical solutions across black-hole and initial-data parameter sets. At inference, each new configuration only requires one cheap numerical-prior evolution, interpolation to the output grid and one FNO forward pass, without per-configuration neural-network retraining.

Methodology

The reproduction uses Patel et al.’s network architecture, loss weights and optimiser schedule. The enhanced PINN extends the L-BFGS optimisation and applies residual-greedy sampling, which places additional training points where the differential-equation residual is largest.

The Schwarzschild hFNO combines a reduced-resolution finite-difference solution with a learned correction. Two coarse solutions, computed with grid factors $k = 2$ and $k = 4$, are combined by Richardson extrapolation to estimate the field that would be obtained at higher resolution [6]. The difference between this estimate and the upsampled $k = 4$ field defines the correction learned during training. The fine finite-difference field is reserved for evaluation and does not enter the training loss.

At inference, the reconstruction is

$$\Phi^{\text{hyb}}(x, t) = \mathcal{U}\Phi^c(x, t) + g(x, t) \mathcal{H}_\theta(X)(x, t). \quad (1)$$

Here $\mathcal{U}\Phi^c$ is the numerical prior obtained by interpolating the $k = 4$ field to the output grid, $\mathcal{H}_\theta$ is the FNO correction [7–9], and g concentrates that correction where the numerical prior varies most rapidly. The training-only $k = 2$ solve and Richardson construction are absent at inference. Each new configuration therefore only requires one coarse $k = 4$ numerical evolution, interpolation and one FNO forward pass. Coarsening space and time by four reduces the finite-difference point-update count to approximately one-sixteenth of the fine evolution, dramatically reducing computational costs at inference.

The Kerr hFNO applies the same numerical-prior-plus-correction principle to a radial Teukolsky evolution of the $(4, 2, 0)$ mode on a horizon-penetrating hyperboloidal domain [10,11]. Its correction is learned from independently evolved lower-resolution fields and is left ungated to preserve the low-amplitude signal at future null infinity. The angular separation constant is fixed at the corresponding Leaver frequency, so the recovered frequency measures reconstruction of the prescribed radial mode rather than an independent spectral prediction. QNMs are extracted using several real and complex estimators over scanned fit windows, with the complete estimator dependence reported in the main report.

Key Findings

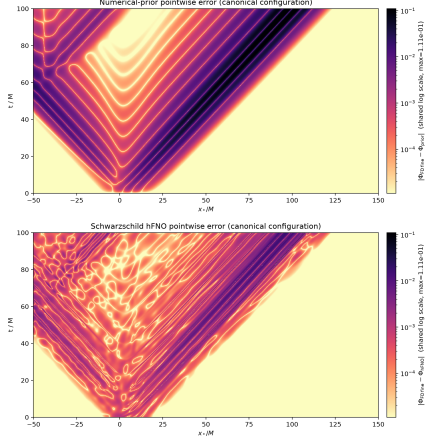


Figure 1: Pointwise Schwarzschild field error on the canonical configuration for the numerical prior (top) and hFNO (bottom), shown on the same logarithmic scale. The hFNO substantially reduces the coherent numerical-prior error bands, while structured wavefront and late-time errors remain.

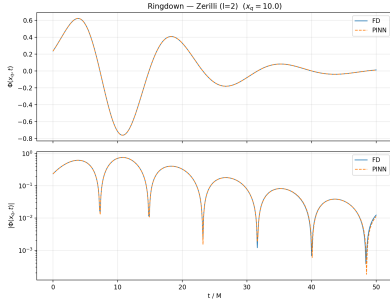


Figure 2: Enhanced PINN ringdown at $x_q = 10M$, showing agreement with the finite-difference reference on linear and logarithmic scales.

Our reproduction obtains relative L^2 field errors of 2.61% for Zerilli and 2.67% for Regge–Wheeler, compared with the published 28.06% and 23.59%. The enhanced PINN reaches 0.58% relative L^2 field error. From its observer waveform, a two-mode fit-window scan gives a 0.029% fundamental-frequency error, while a single-mode nonlinear fit gives a 1.99% damping-time error against Leaver [12].

Across 100 independent Schwarzschild configurations, the hFNO reduces the mean relative L^2 error from 13.39% for the numerical prior to 1.85%, and the mean-squared field error by a factor of 46. The two-mode fit-window method gives a median fundamental QNM frequency error of 0.152% against Leaver. Figure 1 shows that the hFNO substantially reduces the coherent wavefront-error bands of the numerical prior, while structured wavefront and late-time errors remain.

Across 128 independent Kerr configurations spanning $a/M \in [0, 0.95]$, the hFNO achieves a median relative L^2 field error of 0.78% against the $N = 801$ reference and a median frequency error of 0.33% against Leaver. The median damping-time error against Leaver is 1.52%, consistent with the greater sensitivity of the low-amplitude late-time signal. Complete per-estimator results show substantial method dependence.

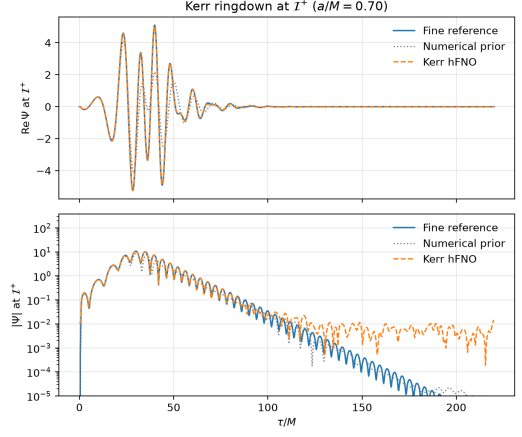


Figure 3: Kerr ringdown at future null infinity for a test configuration nearest $a/M = 0.70$. The Kerr hFNO follows the fine radial reference through the prompt response and early ringdown, whereas the numerical prior is visibly under-resolved. The logarithmic panel reveals the hFNO’s late-time residual, which limits damping-time recovery.

Recommendation & Next Steps

The first priority is a late-time-sensitive training objective that improves the low-amplitude Kerr ringdown. Further tests should cover additional harmonics and initial data, parameters outside the training ranges, and changes of numerical domain and resolution. End-to-end timing should quantify the wall-clock reduction and training break-even point, including numerical evolution, interpolation and neural inference. The resulting waveform families can then be combined with reduced-order likelihood methods to test their practical value for Bayesian parameter estimation [13].

Research Impact & Conclusion

The Schwarzschild and Kerr hFNOs show that a reduced-resolution numerical evolution can retain the governing propagation, scattering and boundary treatment while a neural operator corrects its discretisation error across a parameter family. At inference, each new configuration requires one cheap numerical evolution, interpolation and one FNO forward pass, without neural-network retraining. The resulting field errors of 1.85% and 0.78%, together with frequency errors of 0.152% and 0.33% against Leaver, demonstrate accurate, lower-cost ringdown modelling across the tested parameter spaces. The 1.52% Kerr damping-time error identifies the low-amplitude late-time ringdown as the principal target for improvement.

References

- [1] C. V. Vishveshwara, *Nature* **227**, 936–938 (1970).
- [2] E. Berti, V. Cardoso, and A. O. Starinets, *Class. Quantum Grav.* **26**, 163001 (2009).
- [3] LIGO Scientific Collaboration and Virgo Collaboration, *Phys. Rev. Lett.* **116**, 221101 (2016).
- [4] M. Isi and W. M. Farr, arXiv:2107.05609 [gr-qc] (2021).
- [5] N. Patel, A. Aykatalp, and P. Laguna, arXiv:2401.01440 [gr-qc] (2024).
- [6] L. F. Richardson, *Phil. Trans. R. Soc. A* **210**, 307–357 (1911).
- [7] Z. Li et al., International Conference on Learning Representations (2021).
- [8] K. Um et al., *Adv. Neural Inf. Process. Syst.* **33**, 6111–6122 (2020).
- [9] D. Kochkov et al., *Proc. Natl. Acad. Sci. USA* **118**, e2101784118 (2021).
- [10] S. A. Teukolsky, *Astrophys. J.* **185**, 635–648 (1973).
- [11] E. Harms, S. Bernuzzi, and B. Brügmann, *Class. Quantum Grav.* **30**, 115013 (2013).
- [12] E. W. Leaver, *Proc. R. Soc. Lond. A* **402**, 285–298 (1985).
- [13] P. Canizares et al., *Phys. Rev. Lett.* **114**, 071104 (2015).